# Appendices

# Appendix: Pipeline Stage Reference

# Preparation and analysis I

Table 1: Single-project DAG reference from default `pipeline.yaml` (names shown in dependency order). Scripts live under `scripts/`.

| Stage name | Script / method | Primary inputs | Outputs / artefacts | Failure mode |
|---|---|---|---|---|
| Clean Output Directories | `_run_clean_outputs` | prior `projects/<name>/output/`, mirrored `output/<name>/` targets | emptied trees | Blocking |
| Environment Setup | `scripts/pipeline/stage_00_setup.py` | toolchain probes | scaffold dirs, env exports | Blocking |
| Infrastructure Tests | `scripts/pipeline/stage_01_test.py --infra-only --infra-scope pipeline-smoke` | `tests/infra_tests/` | smoke-test results | Required smoke tests must pass |
| Project Tests | `scripts/pipeline/stage_01_test.py --project-only` | `projects/<name>/tests/` | coverage artefacts | Nonempty tests, own coverage, verifier, and runner |
| Project Analysis | `scripts/pipeline/stage_02_analysis.py` | thin scripts | `figures/`, `data/`, reports | Blocking |

# Search, provenance and rendering I

| Stage name | Script / method | Primary inputs | Outputs / artefacts | Failure mode |
|---|---|---|---|---|
| Connector Search | `scripts/pipeline/stage_08_connector_search.py` | `manuscript/config.yaml` | `output/data/connector_search/` | opt-in (`science` tag); fresh policy-bound skip if unconfigured |
| Provenance Record | `scripts/pipeline/stage_09_provenance_record.py --stage "Connector Search"` | prior stage outputs | `.provenance/graph.json` | opt-in (`provenance` tag); fresh policy-bound skip if unconfigured |
| PDF Rendering | `scripts/pipeline/stage_03_render.py` | `manuscript/`, placeholders | `.pdf`/`.tex` bundles | Blocking |
| Output Validation | `scripts/pipeline/stage_04_validate.py` | render tree | Markdown + PDF diagnostics JSON | Enabled formats, bookends, and provenance block; prose diagnostics advisory |

# Review and delivery I

| Stage name | Script / method | Primary inputs | Outputs / artefacts | Failure mode |
|---|---|---|---|---|
| LLM Manuscript Review | `scripts/pipeline/stage_06_llm_review.py --reviews-only` | resolved manuscript artefacts | textual reviews | Fresh policy-bound skip; required live acceptance remains blocking |
| LLM Translations | `scripts/pipeline/stage_06_llm_review.py --translations-only` | abstract metadata | multilingual snippets | Fresh policy-bound skip; required live acceptance remains blocking |
| Copy Outputs | `scripts/pipeline/stage_05_copy.py` | validated tree | mirrored `output/<name>/…` | Blocking |

# Optional packaging and publication I

| Stage name | Script / method | Primary inputs | Outputs / artefacts | Failure mode |
|---|---|---|---|---|
| Ebook Generation | `scripts/pipeline/stage_11_ebook.py` | rendered combined markdown | `output/ebook/` (EPUB/MOBI/DOCX) | opt-in (`ebook` tag); blocking when invoked |
| docxplus Export | `scripts/pipeline/stage_13_docxplus.py` | project source tree + metadata | `output/docxplus/` (.docx, .docxplus) | opt-in (`docxplus` tag); blocking when invoked |
| Metadata Package | `scripts/pipeline/stage_12_metadata.py` | `manuscript/config.yaml` | `output/metadata/` (ONIX/JSON/OPF) | opt-in (`metadata` tag); blocking when invoked |
| Executable Bundle | `scripts/runner/bundle_executable.py` | project tree + outputs | container bundle manifest | opt-in (`bundle` tag) |
| Archival Publication | `scripts/runner/archive_publication.py` | bundle + deliverables | archival deposit manifest | opt-in (`archival` tag) |

## Optional packaging and publication I

`scripts/pipeline/stage_07_executive_report.py` provides separate **multi-project executive reporting**, not an additional single-project DAG node. `execute_multi_project.py` writes aggregate summaries after project execution; this does not imply automatic invocation of that separate Stage 07 script.

Opt-in stages also prevent completion when invoked and unsuccessful. An allowed skip requires a fresh policy-bound reason receipt; it is not a successful execution. The executable DAG and its contracts remain authoritative.